Python Programming Assignment: Fundamentals

Topics: I/O Functions, Variables, & Operators

Part 1: Input & Output Functions (print & input)

1. Write a program to print "Hello, World!" and your name on two separate lines.
2. Ask the user for their favorite color using input() and print "Your favorite color is [color]".
3. Use a single print() statement to display three different words separated by a hyphen (-).
4. Prompt the user for their birth year and print their age (assume the current year is 2026).
5. Print the result of 5 + 5 such that the output is: The sum of 5 and 5 is 10.
6. Use the end parameter in print() to join two separate print statements with a space.
7. Write a program that takes two strings from the user and prints them joined together.
8. Create a greeting that takes a user's name and prints "Welcome, [Name]!" in all uppercase.
9. Ask for a user's city and country, then print them in the format: "City, Country".
10. Experiment: What happens if you try to add a string and an integer in a print statement? Write a code snippet that fixes this using str().

Part 2: Variables & Data Types

Focus: Direct Assignment and User Input Assignment for Int, Float, String, Complex, and Boolean.

11. Create an integer variable age and a float variable height. Print their types.
12. Store the value 3 + 4j in a variable. Print the variable and its type.
13. Create a boolean variable is_python_fun and set it to True.
14. **Method 1:** Assign three different values to three variables in a single line.
15. **Method 2:** Assign the same value to three different variables in a single line.
16. Take a numeric input from a user and convert it to a float.
17. Take a string input "100" and convert it to an int.
18. Create a variable with a complex number and print only its real part.
19. Define a string variable containing a paragraph and print its length.
20. Swap the values of two variables a and b without using a third variable.

Part 3: Arithmetic Operators

21. Write a program to calculate the area of a rectangle (Length × Width).
22. Take two numbers and print the result of the first raised to the power of the second (a^b).
23. Demonstrate the difference between / (division) and // (floor division) with the numbers 10 and 3.
24. Use the modulus operator % to find the remainder when 25 is divided by 4.
25. Calculate the average of five numbers entered by the user.
26. Create a program that converts minutes into hours and remaining minutes.
27. Calculate the area of a circle where Area = \pi r^2 (Use 3.14 for \pi).
28. Find the cube of a number entered by the user.
29. Perform the calculation 10 + 5 * 2. Does Python follow PEMDAS? Prove it with code.
30. Write a program to calculate simple interest: (P \times R \times T) / 100.

Part 4: Comparison & Logical Operators

31. Compare two numbers entered by the user and print if the first is greater than the second.
32. Check if a user-entered number is even (Number % 2 == 0) and print the Boolean result.
33. Write a program that checks if a number is between 10 and 50 (inclusive) using and.
34. Check if a string entered by the user is equal to "Python".
35. Use the or operator to check if a user is either "Admin" or "Superuser".
36. Demonstrate the not operator by reversing a Boolean variable.
37. Compare two floating-point numbers: 0.1 + 0.2 == 0.3. Explain the result.
38. Take a user's age and check if they are NOT under 18.
39. Check if a number is positive and odd using logical operators.
40. Compare the lengths of two strings provided by the user.

Part 5: Assignment Operators with Arithmetic

41. Initialize a variable x = 10. Use += to add 5 to it.
42. Use -= to subtract 3 from a variable price.
43. Multiply a variable balance by 2 using the *= operator.
44. Divide a variable total by 4 using the /= operator.
45. Use **= to square a variable.
46. Create a counter variable and increment it by 1 using assignment operators.
47. Use %= to find the remainder of a variable divided by 2 and update the variable.
48. Use //= to perform floor division on a variable and update it.
49. Start with n = 2. In three lines of code using assignment operators, turn it into 20.

50. Ask the user for a number, then use += to add 10 to it and print the final result.

Pro-Tip for Students:

Always remember that input() returns data as a **string** by default. If you need to do math, don't forget to wrap it in int() or float()!